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Original Article

AAVrh.10 delivery of novel APOE2-Christchurch variant suppresses amyloid and Tau pathology in Alzheimer's disease mice

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Gene therapy to treat hereditary disorders conventionally delivers the normal allele to compensate for loss-of-function mutations. More effective gene therapy may be achieved using a gain-of-function variant. We tested the hypothesis that AAVrh.10-mediated CNS delivery of the human APOE2 allele with the Christchurch mutation (R136S) (E2Ch) will provide superior protection against APOE4-associated Alzheimer's disease (AD) in mice compared to the unmodified APOE2 allele (E2). The vectors were assessed in two mouse strains with humanized APOE4: APP.PSEN1/TRE4 "amyloid mice" and P301S/TRE4, "tau mice." Both the E2Ch and E2 vectors prevented Aβ42 and Aβ40 accumulation and decreased β-amyloid aggregates in amyloid mice, but only the E2Ch vector suppressed tau tangles in tau mice. Microglial activation and reactive astrocytes were significantly suppressed by both vectors in amyloid mice but only the E2Ch vector mediated significant suppression of Iba1 and glial fibrillary acidic protein (GFAP) in tau mice. In four behavioral assays, the E2 and E2Ch vectors had similar benefits in amyloid mice, but E2Ch outperformed E2 in tau mice. In summary, while E2 is effective in suppressing amyloid pathology, the novel E2 variant E2Ch more effectively treats both the amyloid and tau pathology of murine AD in APOE4 background, supporting the development of AAVrh.10APOE2Ch as a therapy for APOE4-associated AD.

INTRODUCTION

Gain-of-function mutations result in the gene product having enhanced and/or sometimes altered function.[1–5] While many naturally occurring gain-of-function variants in the human genome are associated with increased risk for disease,[6,7] there are rare examples (such as the Padua variant of factor IX) where a human gain-of-function mutation has been used as highly effective therapy to treat disease.[8] The focus of this study is to evaluate the combination of the rare Christchurch (R136S) variant of apolipoprotein E (APOE) with the neuroprotective APOE2 (C112/C158) variant to generate a highly effective gain-of-function variant that can be delivered to the CNS by adeno-associated virus (AAV)-mediated gene therapy to suppress the development of both amyloid and tau-associated pathology and associated CNS dysfunction in murine models of APOE4 associated Alzheimer's disease (AD).

APOE, is a299-amino-acid lipid transport protein that functions as a major carrier of cholesterol in the CNS[9,10] and is the major genetic risk factor for sporadic late-onset AD.[11–15] APOE has three common variants: APOE3 (C112/R158), APOE4 (R112/R158), and APOE2 (C112/C158).[10] There are extensive epidemiologic, clinical, and experimental data demonstrating that APOE3 is associated with an average risk for the development of Alzheimer's disease, while APOE4 conveys a high risk and APOE2 is protective.[16–31] These data, and the observation that, in E2E4 heterozygotes, inheritance of E2 cancels out most of the deleterious effects of the E4 allele,[20,21,23,26] led to the hypothesis that gene-therapy-mediated expression of APOE2 in the CNS of APOE4 homozygotes would biologically modify the E4 brain to that of an E2E4 heterozygote, reducing the risk for AD.[32–35] This concept is being tested in humans in a clinical trial with AAV serotype rh.10 delivery of APOE2 to the CNS of APOE4 homozygotes with early-onset AD (clinicaltrials.gov NCT03634007).

All of these variants are in the 112–158 region of the APOE gene, a region that codes for binding to the low-density lipoprotein (LDL) receptor and heparan proteoglycans[36] (Figure 1A). The importance of the APOE 112–158 region relating to AD was highlighted by the case report of a female carrier of the PSEN1/E280A mutation who was homozygote for APOE3Christchurch (Ch), the R136S variant superimposed on the average risk APOE3 variant.[37] The dominant PSEN1/E280A gene defines a Colombian family of >1,200 individuals all with early-onset CNS amyloid accumulation, with heterozygotes developing dementia in their mid-forties.[38,39] Strikingly, the female who co-inherited homozygous APOE3Ch and heterozygous PSEN1-E280A was in good cognitive condition at age 70 years, well past the age when the kindred typically develop cognitive decline.[37]

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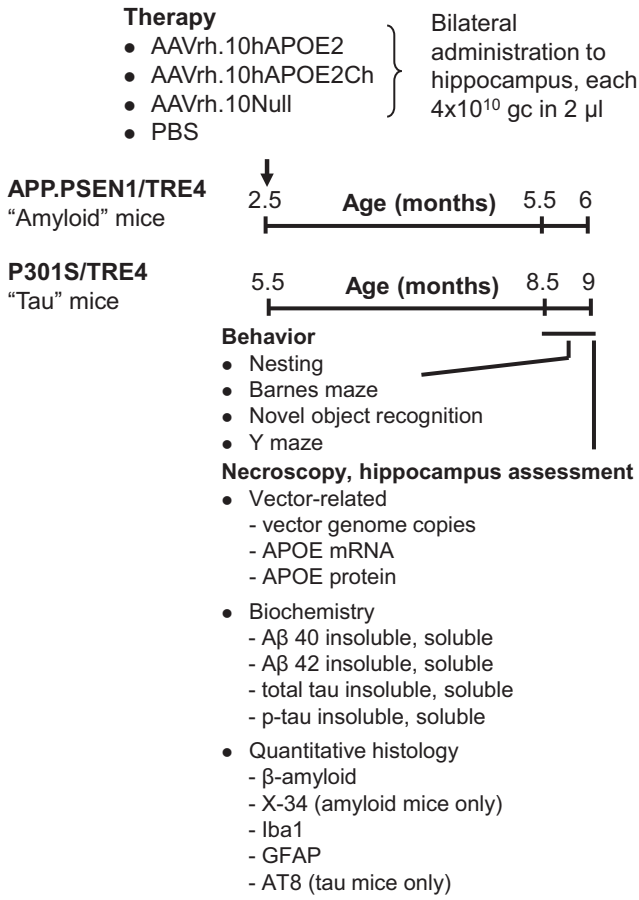


Figure 1. Experimental design for comparison of treatment and controls in amyloid and tau mouse models

The amyloid mice were treated at 2.5 months and assessed at 5.5–6 months (behavior) and sacrificed at 6 months for vector-related, biochemical, and histologic parameters in the hippocampus. The tau mice were treated at 5.5 months and assessed at 8.5–9 months (behavior) and sacrificed at 9 months for the hippocampal parameters. The numbers of mice assessed for each parameter are detailed in Figures 3, 4, 5, 6, and 7 and Table S5.

Clinical assessment revealed accumulation of amyloid plaques, but low levels of tau burden, leading to the concept that the Christchurch variant functioned as a gain-of-function allele that prevents the tau component of AD pathology. In a subsequent study, Almeida et al.[40] demonstrated in a single-cell study the protective effects of the Ch variant. This finding was further supported by Quiroz et al.,[41] where the Ch variant was linked to the delayed onset of cognitive impairment in individuals who have at least one Ch allele in the PSEN1/E280A kindred. *In vitro* experiments demonstrated that APOE3Ch behaves like the APOE2 allele; both bind poorly to heparin (binding, APOE2 > APOE3Ch) compared to APOE4, which binds tightly.[37] Research involving both humans and animals has established a link between APOE, heparan sulfate proteoglycans (HSPGs), and the pathobiology of AD, suggesting that their interactions may be essential for the disease's progression. APOE4, a prominent risk factor for sporadic AD, binds more strongly to HSPGs compared to the protective variants APOE2, APOE3Ch, and APOE3 V236E, which show weaker binding, as validated in study by Quiroz et al.[41] Additionally, inhibiting glycosaminoglycans, including HSPGs, either genetically or competitively, has been shown to limit the spread of neurotoxic tau seeds and influence amyloid-β dynamics *in vitro* and in AD mouse models.[42,43] Therefore, modulating the interactions between APOE and HSPGs may be vital for enhancing resistance to neurodegeneration linked to AD, which might be one of the major mechanisms in protective effect of APOE3Ch mutation. Subsequent to the original case report, additional support that the Christchurch is a protective gain-of-function APOE variant has come from several sources: (1) a second similar case with a similar clinical course has been described[44]; and (2) 37 Christchurch heterozygotes among 455,306 UK Biobank participants were shown to be free of AD.[27] Further, experimental studies have identified possible mechanisms by which the Christchurch variant conveys protection against AD.[45–47]

Our study is based on the observation that inheritance of APOE3Ch prevented the PSEN1/E280A-driven development of tau pathology but not amyloid pathology,[37] together with our prior gene therapy study demonstrating that APOE4-associated amyloid pathology can be suppressed by AAVrh.10-mediated delivery of APOE2 to the CNS.[48] Therefore, we sought to assess whether gene delivery of AAVrh.10hAPOE2Ch (the combination of the Christchurch variant [R136S] with the APOE2 [C112/C158] variant) would be more efficacious than the APOE2 alone in suppressing amyloid and p-tau pathology and consequent neurodegeneration associated with AD (Figure 1). We tested the consequences of direct hippocampal administration of AAVrh.10hAPOE2Ch (E2Ch) compared to AAVrh.10hAPOE2 (E2) on Alzheimer's disease-related pathology and behavior in two murine models: APP.PSEN1/TRE4 mice (mutant human presenilin-1 mutation with APOE4 replacement, amyloid mice) and P301S/TRE4 mice (human P301S mutation with APOE4 replacement, tau mice). We hypothesize that E2Ch and E2 vectors will (1) similarly suppress the amyloid pathology in amyloid mice and (2) while the E2 vector alone may not suppress the p-tau pathology, the addition of the Christchurch variant to the E2 variant functions effectively to suppress p-tau pathology in the tau mice; i.e., the Christchurch mutation is synergistic with APOE2, functioning as a highly effective gain of function to treat both the amyloid and p-tau pathology of AD. If the data support these hypotheses, E2Ch would be a candidate as a second-generation AAV therapy to treat APOE4 homozygote-associated AD.

RESULTS

Novel APOE2Ch showed decreased heparin-binding affinity relative to APOE3Ch and APOE2

APOE predominantly binds and interacts with heparin through residues 136–150 (Figure 2A). Like Arboleda-Velasquez et al.,[37] our results show that APOE4 has the highest affinity for heparin, APOE3 intermediate, APOE2 a lower affinity, and APOE3Ch the lowest. Further, we observed that APOE2Ch binding is even lower than the previous lowest-binding variant, APOE3Ch (Figure 2B).[37,49]

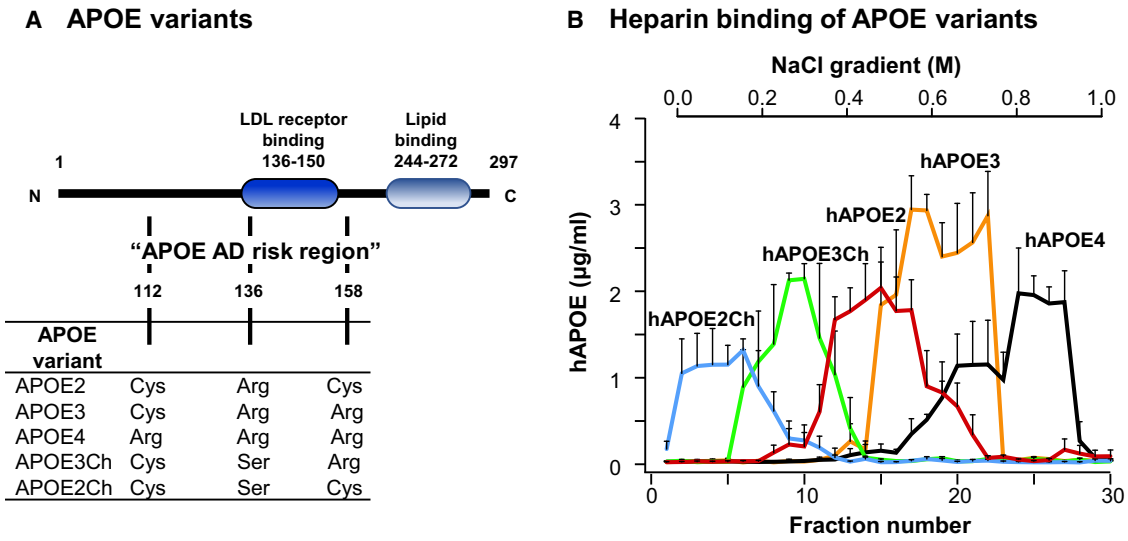


Figure 2. Comparison of the binding efficacy of the APOE variants to heparin

(A) APOE variants. The APOE AD risk region including the Columbian APOE3Ch variant and the therapeutic gain-of-function APOE2Ch variant to be compared to the APOE2 variant in the amyloid and tau murine models. (B) Heparin-binding results of the APOE variants. Data are from three independent experiments and represented as mean ± SEM.

The relevance of this binding phenomenon has been demonstrated by Marino et al.,[43] where they show that E3Ch mimetic antibody blocks the APOE-HSPG interaction responsible for alleviated APOE-induced neurotoxicity.

Intrahippocampal delivery of AAVrh.10hAPOE2 and AAVrh.10hAPOE2Ch results in comparable levels of hAPOE2 and hAPOE2Ch expression

To ensure all experimental groups had comparable AAV transduction, vector genome copies, APOE mRNA, and APOE protein were assayed in all study cohorts and controls in the hippocampus of both the amyloid and tau mice (Figure 3). No vector genome copies were detected in the PBS cohort in either the amyloid or tau mice (Figures 3A and 3B) and there were no significant differences in vector copies in null (identical AAV with no translatable expression cassette), E2-, and E2Ch-treated amyloid ($p > 0.1$, Figure 3A) or tau mice ($p > 0.2$, Figure 3B). No significant difference in APOE mRNA levels were observed between controls PBS and null groups in amyloid ($p > 0.5$, Figure 3C) or tau mice ($p > 0.9$, Figure 3D). Treatment with E2 and E2Ch each had significantly greater mRNA levels in both amyloid and tau mice compared to PBS ($p < 0.01$, Figure 3C) and null ($p < 0.01$, Figure 3D) controls. PBS and null groups had similar background levels of hAPOE protein in amyloid and tau mice ($p > 0.9$, Figures 3E and 3F). The E2- and E2Ch-treated mice had comparable ($p > 0.9$) 11- to 27-fold higher hAPOE protein levels in amyloid and tau mice ($p < 0.01$, Figures 3E and 3F) than background. In addition to the administered region (hippocampus), APOE protein levels assessed in cortex, midbrain, and cerebellum (Table S1) were elevated at levels of 6.5- to 28-fold over the endogenous human APOE level of TRE4 mice compared to 26-fold in the hippocampus.

E2 and E2Ch treatment decreased Aβ42 and Aβ40 levels in hippocampus of amyloid mice

The effect of treatments on amyloid levels were investigated with quantification of hippocampus Aβ42 and Aβ40 levels in insoluble and soluble fractions (Figures 4A and 4B). In amyloid mice, no significant difference between PBS and null groups were observed in insoluble or soluble fractions of Aβ42 ($p > 0.3$ insoluble, $p > 0.9$ soluble, Figure 4A) or Aβ40 ($p > 0.3$ insoluble, $p > 0.9$ soluble, Figure 4B). E2 treatment significantly suppressed insoluble and soluble Aβ42 levels compared to PBS and null groups by 79.7% ($p < 0.01$) and 80.0% ($p < 0.01$), respectively, and Aβ40 levels by 68.0% ($p < 0.01$) and 68.1% ($p < 0.01$), respectively (Figures 4A and 4B). Additionally, E2Ch significantly decreased insoluble and soluble Aβ42 levels by 80.5% ($p < 0.01$) and 81.6% ($p < 0.01$) and Aβ40 levels by 66.6% ($p < 0.01$) and 71.0% ($p < 0.01$) compared to PBS and null groups, respectively. No significant difference was observed between E2 and E2Ch treatment in both Aβ42 and Aβ40 levels in insoluble and soluble fractions ($p > 0.8$ all comparisons; Figures 4A and 4B). Next, to understand the impact of the APOE2 expression, we assessed the regional Aβ levels in amyloid mice. The data in Table S2 demonstrate the correction extends beyond the administration site in the hippocampus into cortex, midbrain, and cerebellum. The extent of correction was similar in all regions tested.

E2Ch treatment diminished tau burden

Sequential extraction from hippocampal tissue was used to quantify total tau and p-tau levels in each cohort (Figures 4C and 4D). The levels of total tau (Figure 4C) and p-tau (Figure 4D) were similar in the PBS and null groups. In contrast, total tau in the insoluble

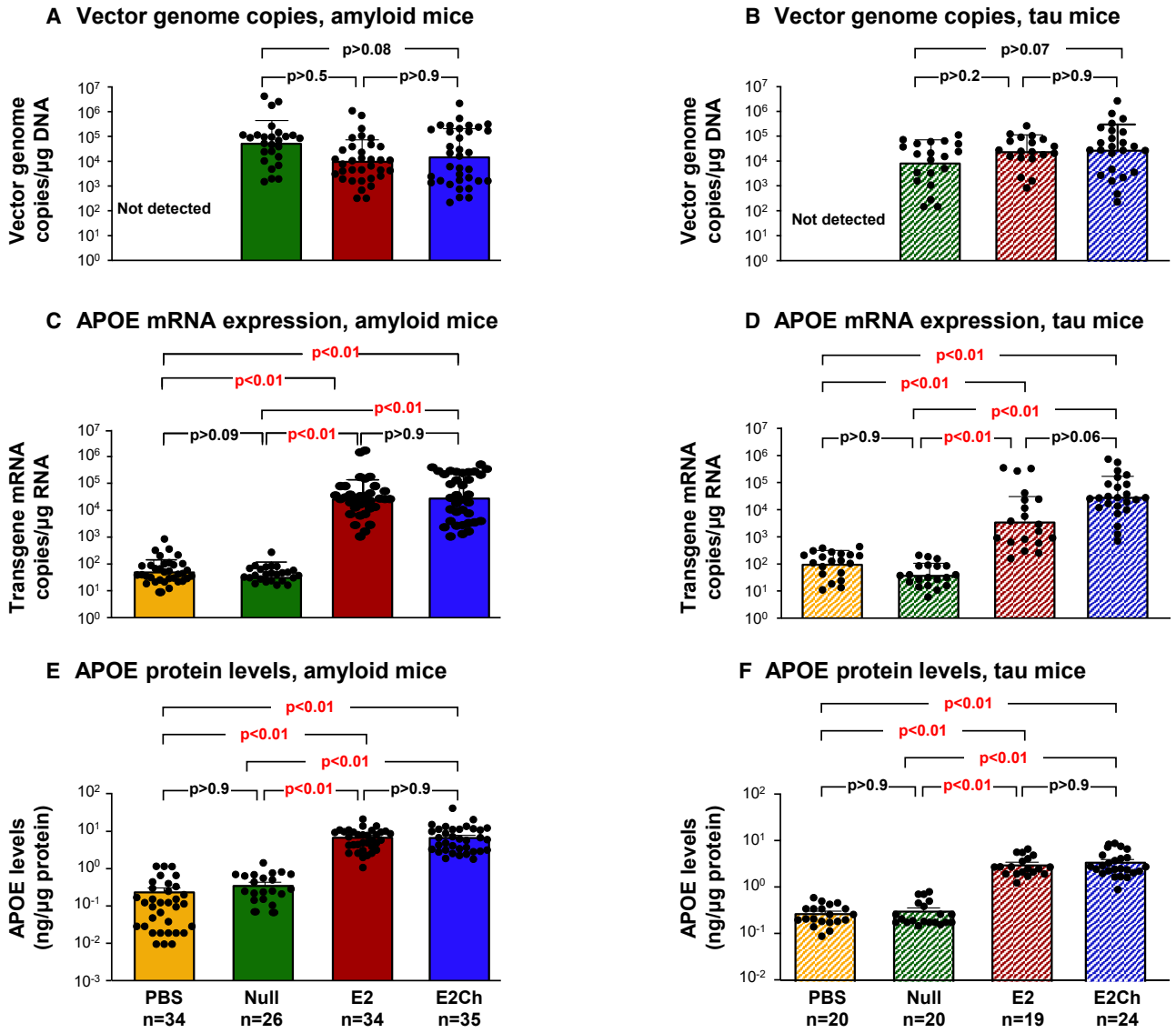


Figure 3. Vector-mediated expression in the hippocampus of amyloid and tau mice

(A and B) Vector genome copies (gc), amyloid and tau; (C and D) APOE mRNA, amyloid and tau mice; (E and F) APOE protein levels, amyloid, and tau mice. Data are presented as mean ± standard error; the "n" is the number of mice and each dot represents an individual mouse.

fraction was significantly lower in E2 group ($p < 0.01$ compared to PBS) but not compared to null ($p > 0.1$, Figure 4C). Soluble tau levels were unchanged in the E2 cohort (compared to PBS, and to null, Figure 4C) as were the insoluble and soluble fractions of p-tau (compared to PBS in both fractions, and to null, Figure 4D). In contrast, administration of E2Ch decreased total tau levels in the insoluble fraction by 40.5% and in the soluble fraction by 68.6% compared to PBS ($p < 0.01$ both insoluble and soluble fractions) and compared to null ($p < 0.01$ insoluble, $p < 0.05$ soluble, Figure 4C). Additionally, E2Ch significantly decreased p-tau levels by 36.2% in the insoluble fraction and 44.5% in the soluble fraction compared to PBS group ($p < 0.01$ insoluble, $p < 0.05$ soluble, Figure 4D) and compared to null ($p < 0.01$ insoluble, $p < 0.01$ soluble, Figure 4D) controls. Importantly, a significant difference was observed between the E2Ch and E2 treatments for the amounts of soluble total tau ($p < 0.01$, Figure 4C), soluble p-tau ($p < 0.05$, Figure 4D) and insoluble p-tau ($p < 0.05$, Figure 4D), indicating a pronounced effect of the E2Ch greater than E2 on reducing tau pathology. Next, to understand the impact of the APOE2 expression, we assessed the regional tau and phospho-tau levels in the tau mice. The data in Table S3 demonstrate the correction extends beyond the administration site in the hippocampus into cortex, midbrain, and cerebellum. The extent of correction was similar in all regions tested.

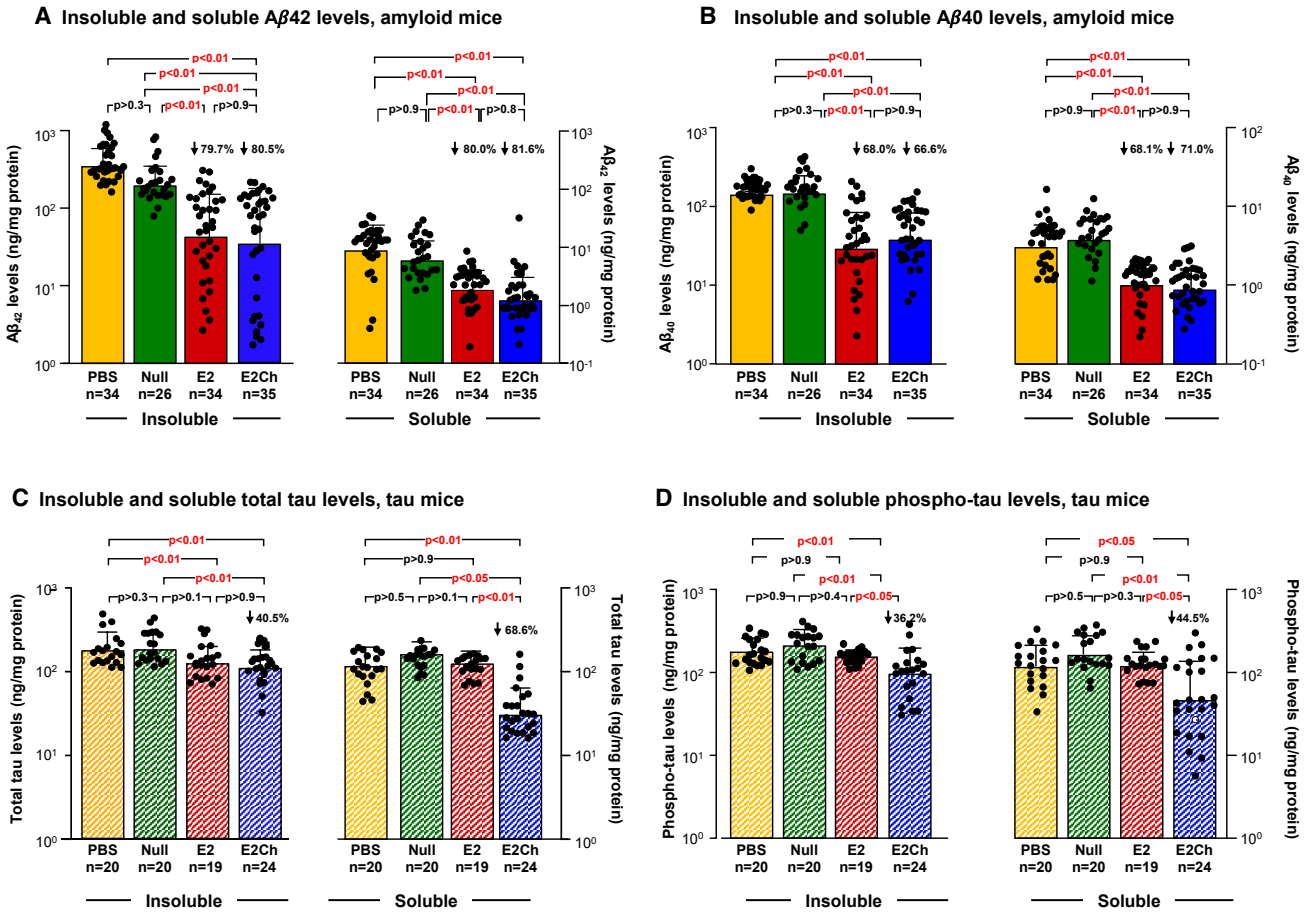


Figure 4. Aβ42 and Aβ40 levels in the hippocampus of amyloid and tau mice treated with E2 vs. E2Ch

(A and B) Insoluble and soluble Aβ42 and Aβ40 amyloid mice. (C and D) Insoluble and soluble total and phosphor-tau tau mice. Data are presented as geometric mean ± standard deviation; each dot represents an individual mouse.

E2 and E2Ch treatment decreased number of amyloid plaques and diminished microgliosis/astrogliosis

For detection of β-amyloid MOAB-2 antibody was used to examine the accumulation of β-amyloid plaques.[50] The PBS and null groups showed similar levels of β-amyloid staining ($p > 0.07$, Figures 5A, 5B, and 5E). Both E2 (Figure 5C) and E2Ch (Figure 5D) treatments led to a significant decrease in MOAB-2 immunopositivity of β-amyloid ($p < 0.01$ for E2 and $p < 0.01$ for E2Ch compared to PBS and to null; $p < 0.01$ for both E2 and E2Ch compared to null, Figure 5E). No significant difference was observed between treatment with E2 and E2Ch ($p > 0.9$, Figure 5E).

X-34 staining was used to evaluate the presence of amyloid aggregates.[51] Similar to the MOAB-2 staining, the PBS (Figure 5F) and null (Figure 5G) groups showed comparable X34 staining ($p > 0.05$, Figure 5J). Treatment with both E2 (Figure 5H) and E2Ch (Figure 5I) resulted in a decrease of X34 staining ($p < 0.01$ for both E2 and E2Ch compared to PBS), demonstrating that both E2 and E2Ch mediated suppression of the accumulation of amyloid aggregates in amyloid mice ($p > 0.4$, Figure 5J).

Iba-1 immunostaining provides a measure of microglial activation.[52] Both the PBS (Figure 5K) and null (Figure 5L) groups displayed strong staining profiles with Iba-1 indicating similar levels of microglial activation. In contrast, treatment with both E2 (Figure 5M) and E2Ch (Figure 5N) significantly inhibited the immunopositivity of Iba-1 ($p < 0.01$ for E2 and E2Ch, Figure 5O) when compared to the PBS and null groups. There was no difference between the E2 and E2Ch groups ($p > 0.9$, Figure 5O). These results suggest that both E2 and E2Ch have similar capacity to suppress the microglial-related neuroinflammation associated with amyloid pathology.

Glial fibrillary acidic protein (GFAP) immunostaining provides an additional measure of neuroinflammation via astroglia activation.[53] Both the PBS (Figure 5P) and null (Figure 5Q) groups displayed strong staining with GFAP indicating similar levels of astroglia

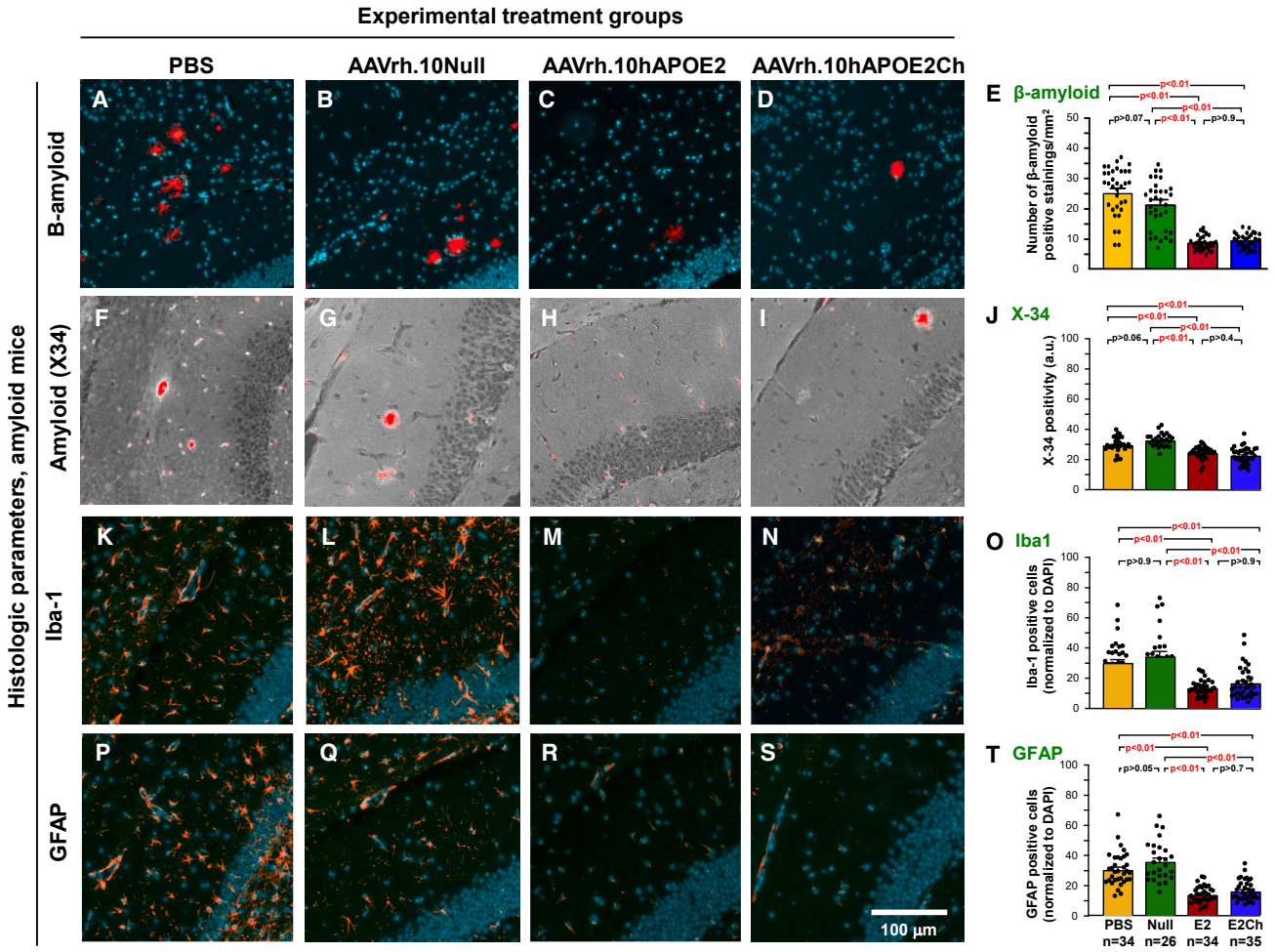


Figure 5. Histologic parameters β-amyloid, X-34, Iba-1, and GFAP in treatment groups compared to the AAVrh.10Null and PBS control treatment groups in the CA-1 region of the hippocampus of amyloid mice

(A–D, F–I, K–N, and P–S) Images to demonstrate treatment effect. (E) Amyloid, Aβ aggregates, (J) X-34, amyloid fibrils, (O) Iba-1, microglial activation, and (T) GFAP, astroglial activation are the quantitative analyses of the histochemistry. Data are presented as mean ± standard error; each dot represents an individual mouse; β-amyloid, Iba-1, and GFAP staining are shown in red. DAPI-stained nuclei are shown in blue. The dentate gyrus (dense collection of nuclei) is visible at the bottom right corner of each panel. For X34 staining, monochrome images are shown. The red color indicates high-intensity staining (staining above a fixed intensity threshold). In X34 panels, the nuclei of the dentate gyrus are visible in the lower right corner of the field as stain-negative ovals. Bar, 100 µm.

activation. However, treatment with both E2 (Figure 5R) and E2Ch (Figure 5S) significantly inhibited the immunopositivity of GFAP ($p < 0.01$ for E2 and E2Ch, Figure 5T) when compared to the PBS and null groups. There was no significant difference between E2 and E2Ch groups ($p > 0.7$, Figure 5T), suggesting that both E2 and E2Ch treatments have the capacity to suppress amyloid-associated astroglia activation.

E2Ch suppressed tau tangle formation and mitigated microgliosis/astrogliosis

AT8 staining detects phosphorylated tau.[54] There was a similar level of staining in both the PBS (Figure 6A) and null (Figure 6B) groups ($p > 0.9$, Figure 6E). E2 did not induce any significant changes in AT8 staining (Figure 6C) compared to the control groups ($p > 0.9$, Figure 6E). However, E2Ch (Figure 6D) significantly suppressed AT8 staining when compared to the PBS, null, and E2 groups ($p < 0.01$, $p < 0.01$, $p < 0.05$, respectively, Figure 6E).

Both the PBS (Figure 6F) and null (Figure 6G) groups had strong staining profiles in the tau mice with Iba-1 indicating similar levels of microglial activation. However, treatment with E2Ch (Figure 6I) significantly inhibited the immunopositivity of Iba-1 when compared to the E2 ($p < 0.01$ Figures 6H and 5J), PBS, and null groups (both $p < 0.01$, Figure 6J), supporting the conclusion that E2Ch treatment is efficacious for mitigating tau pathology-related neuronal inflammation whereas E2 is not.

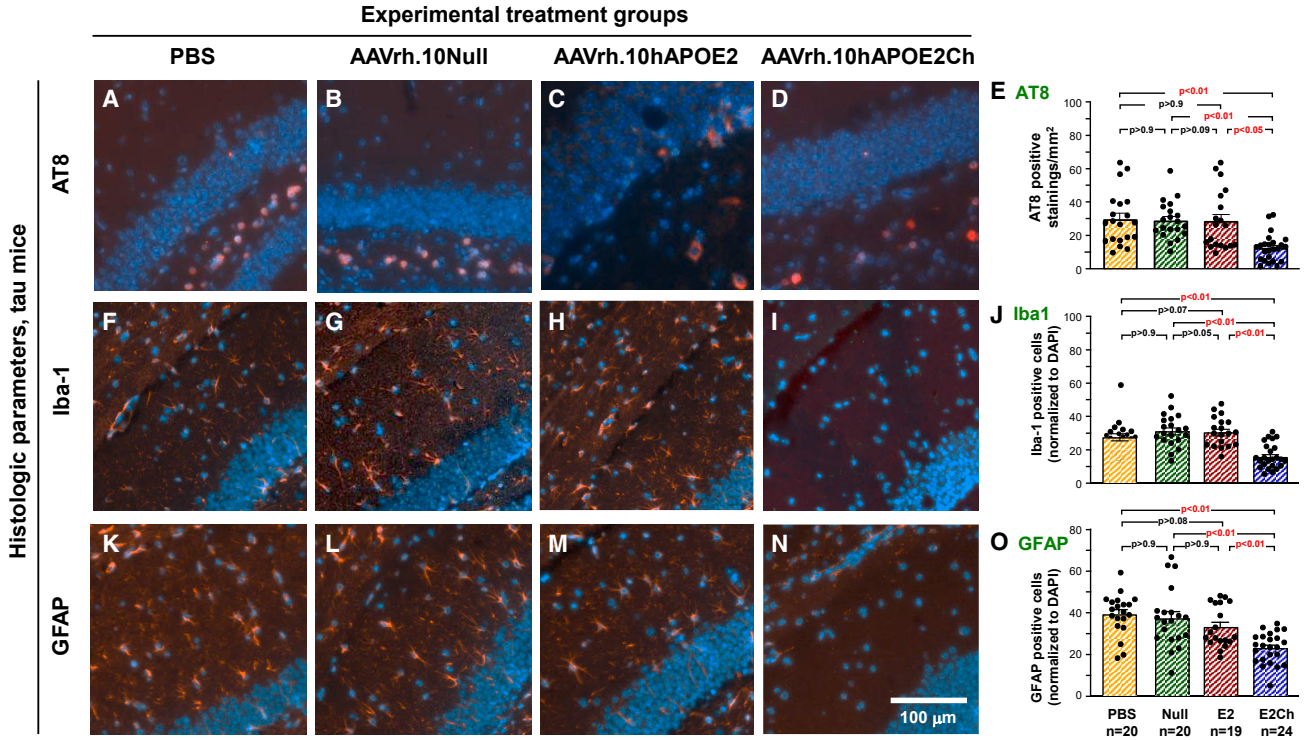


Figure 6. Histologic parameters AT8, Iba-1, and GFAP in treatment groups compared to controls in the CA-1 region of the hippocampus in tau mice

(A–D, F–I, and K–N) Representative examples. (E) AT8, tau burden and aggregates; (J) Iba-1, microglial activation; and (O) GFAP, astroglial activation are the quantitative analyses of the histochemistry. Data are presented as mean ± standard error; each dot represents an individual mouse. AT8, Iba-1, and GFAP staining are shown in red. DAPI-stained nuclei are shown in blue. The dentate gyrus is visible at the bottom right corner of each panel. Bar, 100 µm.

Both the PBS (Figure 6K) and null (Figure 6L) groups displayed strong staining with GFAP indicating similar levels of astroglia activation. However, treatment with E2Ch (Figure 6N) significantly inhibited the immunopositivity of GFAP when compared to the E2 ($p < 0.01$ Figures 6M and 5O), PBS, and null groups (both $p < 0.01$ Figure 6O); i.e., E2Ch treatment is efficacious for mitigating tau pathology-related astroglia activation, whereas E2 is not.

E2Ch significantly altered microglia/astroglia proinflammatory gene expression in amyloid and tau mice

As an independent assessment of the comparative benefits of E2 or E2Ch gene therapy in both amyloid and tau models, we assessed the microglia/astroglia phenotype as indicated by the expression level of inflammatory and anti-inflammatory markers, including Arg-1, CD163, CD68, CLEC7A, TREM2, and CD206 by RT-qPCR.[55] From the data generated with these inflammatory and anti-inflammatory markers, it was clear that the glial cells were less activated in the amyloid E2 and E2Ch cohorts and the tau E2Ch cohorts (Figure S1).

E2Ch alleviated behavioral deficits in amyloid and tau mice

The genetic backgrounds for the E2 and E2Ch mice are unique and do not correspond to the C57Bl/6 or BALB/c standard wild-type mice. As a check on our methods, we assessed C57Bl/6 mice on each of the behavior tests described below and found that, for all tests in our lab, the C57Bl/6 mice behave in the expected age-appropriate way.

The nesting test reflects general well-being, motor coordination, and cognitive function. Impairments in nesting behavior have been observed in animal models of AD.[56] A significant difference was observed between E2 and PBS treatments in amyloid mice ($p < 0.03$), but not between E2 and null, E2Ch and the controls, and E2 and E2Ch (all comparisons $p > 0.9$, Figure 7A). In the nesting test for tau mice, E2Ch, but not E2, treatment resulted in improved nesting score compared to the PBS and null controls ($p < 0.01$), and the E2Ch group outperformed the E2 group ($p < 0.04$, Figure 7B).

The Y-maze test assesses spatial working memory and spontaneous alternation behavior, measuring the ability of the animals to remember and navigate through a series of arm choices in a maze.[57] Impairments in spontaneous alternation have been linked to memory deficits and cognitive decline.[58] To ensure that the Y-maze performance was valid, all groups were tested for total number of entries; no significant difference was observed ($p > 0.9$ for all comparisons among the amyloid mouse groups, Figure S2A; $p > 0.06$ all comparisons among the tau mice groups, Figure S2C). In the Y-maze test for amyloid mice, both E2 and E2Ch treatments improved alternation ratios compared to the PBS ($p < 0.01$ for E2

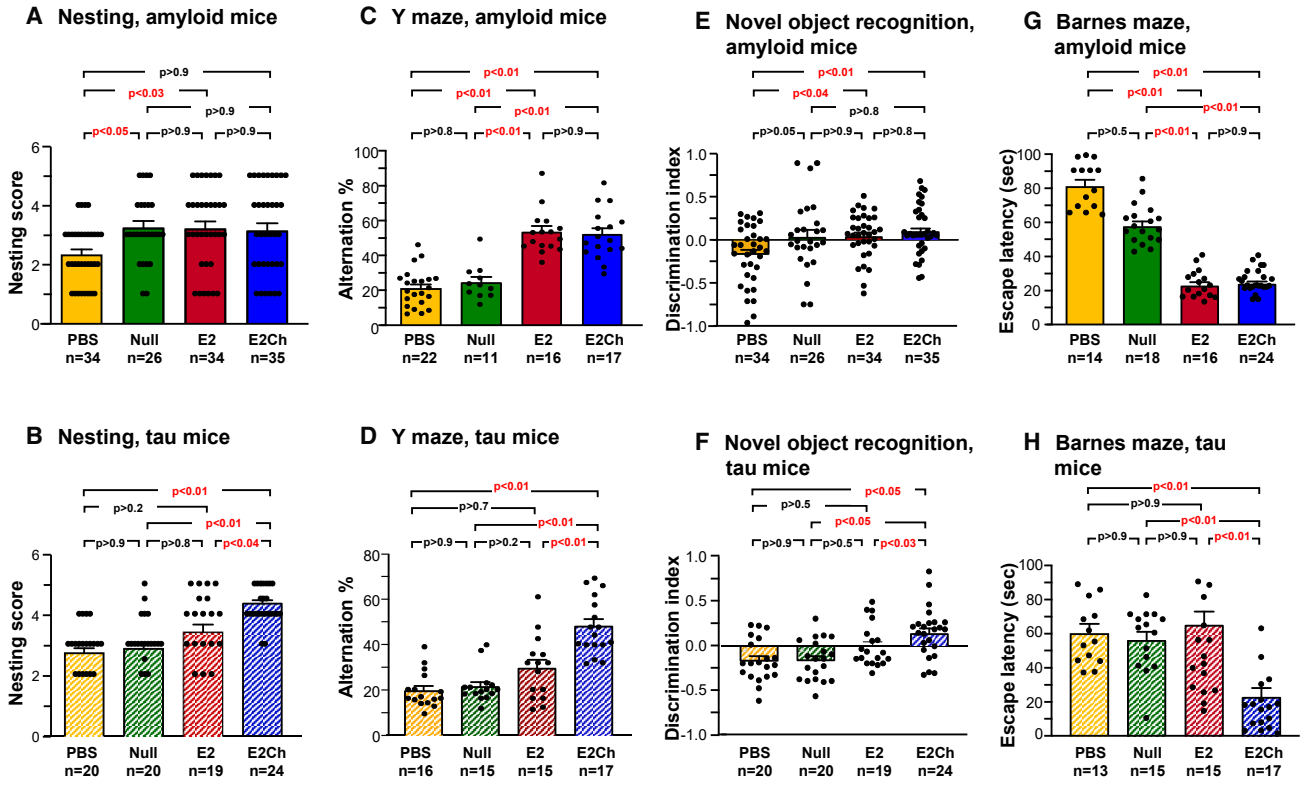


Figure 7. Assessment of effect of treatment and controls on nesting, Y maze, novel-object-recognition, and Barnes maze tests in amyloid and tau mice

(A and B) Nesting, amyloid, and tau mice; (C and D) Y maze, amyloid mice and tau; (E and F) novel-object recognition, amyloid and tau mice; (G and H) Barnes maze, amyloid and tau mice. Data are presented as mean ± standard error; each dot represents an individual mouse.

and E2Ch, Figure 7C) and null groups ($p < 0.01$ for E2 and E2Ch), but no significant difference was observed in the amyloid mice between E2 and E2Ch treatments ($p > 0.9$, Figure 7C). In contrast, in the Y-maze test for tau mice, E2Ch showed significantly higher alternation ratios compared to all other groups ($p < 0.01$ compared to all groups, Figure 7D). Similarly, the assay was valid for tau mice as all groups had indistinguishable number of entries (Figure S2C), but tau mice treated with E2Ch were significantly different from E2 ($p < 0.04$), PBS ($p < 0.02$), and null ($p < 0.03$), whereas there were no differences between E2 and controls (PBS $p > 0.2$, null $p > 0.1$, Figure S2D).

The novel-object-recognition test is employed to evaluate memory and the ability to discriminate between familiar and novel objects.[59] This test assesses the animal's innate preference for novelty and helps to assess memory deficits and impaired recognition. For the novel-object-recognition test in amyloid mice, both E2 and E2Ch treatments showed significantly higher discrimination indices compared to PBS ($p < 0.01$, $p < 0.04$, respectively), with no significant difference between E2 and E2Ch ($p > 0.8$, Figure 7E). However, in the tau mice, E2Ch treatment had significantly higher discrimination indices compared to PBS and null (both $p < 0.05$) and was significantly more efficacious compared to treatment with E2 ($p < 0.03$, Figure 7F).

For the Barnes maze, animals were trained for 7 days for identification of visual cues to locate the escape hole. After 7 days of training, the last 24 h of data were assayed as a measure of performance (Figure S3). In the amyloid mice, the Barnes maze test demonstrated that both E2 and E2Ch significantly decreased escape latency compared to PBS and null (both $p < 0.01$ for E2 and E2Ch, Figure 7G), indicating enhanced spatial learning and memory retention.[60] In the amyloid mice, there was no significant difference between E2 and E2Ch treatments ($p > 0.9$, Figure 7G). In tau mice, no significant effect was observed in E2 groups compared to PBS and null groups ($p > 0.9$). In contrast, in the tau mice, E2Ch treatment significantly reduced escape latency compared to PBS, null, and E2 (all $p < 0.01$, Figure 7H), and E2Ch was significantly better than E2 ($p < 0.01$), consistent with the concept that, relevant to tau pathology, E2Ch treatment is efficacious for enhanced spatial learning and memory retention, whereas E2 is not.

Data summary

In summary, both E2 and E2Ch vectors were equally effective in suppressing APOE4-associated pathology in amyloid mice, whereas only E2Ch was effective in suppressing APOE4-associated pathology in the tau mice (Table 1).

Table 1. Summary of assessments of AAVrh.10APOE2Ch vs. AAVrh.10APOE2 administration on biochemical, histologic and behavioral parameters in APP.PSEN1/TRE4 (amyloid) and P3015/TRE4 (tau) mice[a,b]

	APP.PSEN1/TRE4 (amyloid) mice				P301S/TRE4 (p-tau) mice		
Parameters	E2 > controls[c]	E2Ch > controls[c]	E2Ch vs. E2	Parameters	E2 > controls[c]	E2Ch > controls[c]	E2Ch vs. E2
Biochemical				Biochemical			
Aβ40, insoluble	+	+	–	total tau, insoluble	+	+	–
Aβ40, soluble	+	+	–	total tau, soluble	–	+	+
Aβ42, insoluble	+	+	–	p-tau, insoluble	–	+	+
Aβ42, soluble	+	+	–	p-tau, soluble	–	+	+
Histologic				Histologic			
Amyloid B	+	+	–	AT8	–	+	+
X-34	+	+	–	Iba1	–	+	+
Iba1	+	+	–	GFAP	–	+	+
GFAP	+	+	–				
Behavioral				Behavioral			
Nesting	–	–	–	nesting	–	+	+
Y maze	+	+	–	Y maze	–	+	+
Novel-object recognition	–	–	–	novel-object recognition	–	+	+
Barnes maze	+	+	–	Barnes maze	–	+	+
Summary				Summary			
Biochemical tests	4/4, 100%	4/4, 100%	0/4, 0%	biochemical tests	1/4, 25%	4/4, 100%	3/4, 75%
Histologic tests	4/4, 100%	4/4, 100%	0/4, 0%	histologic tests	0/4, 0%	3/3, 100%	3/3, 100%
Behavioral tests	2/4, 50%	2/4, 50%	0/4. 0%	behavioral tests	0/4, 0%	4/4, 100%	4/4, 100%

[a]Summary of significant parameters based on the data in Figures 3, 4, 5, 6, and 7.
[b]Abbreviations and symbols: E2, AAVrh.10hAPOE2; E2Ch, AAVrh.10hAPOE2Ch; E2Ch vs. E2, direct comparison of a significant difference of the two vectors in each test (see Figures 3, 4, 5, 6, and 7); +, E2Ch significantly more effective than E2;–, E2Ch and E2 similar.
[c]To be considered significant, E2 and E2Ch must be significant ($p < 0.05$) compared to both PBS and null controls.

DISCUSSION

In this study, we confirmed our hypothesis that the Christchurch variant together with the APOE2 isoform generates a "hyper-gain-of-function" APOE that, when delivered to the CNS by AAV-mediated gene therapy, would suppress both APOE4-associated amyloid and tau pathology. Using human APOE4 homozygote amyloid and tau mouse models, we demonstrated that AAVrh.10 coding for E2 or E2Ch similarly improved amyloid pathology and behavioral abnormalities in amyloid mice, while E2Ch, but not E2, remarkably improved tau-associated AD pathology assessed by biochemical, histological, and behavioral parameters. Together, these data support the development of APOE2Ch-based gene therapy for the treatment of APOE4-associated AD.

APOE isoforms and the risk for AD

The human APOE gene has three common alleles: E2, E3, and E4. From 15% to 20% of the population carry at least one copy of E4.[28,29,61,62] Individuals who carry one copy of the E4 allele have an increased risk of developing AD, while those who inherit two copies have a higher risk and develop AD at an earlier age.[16–30,62]

The role of APOE in AD pathology and progression is multifaceted. APOE is produced and secreted in the brain by astrocytes and, to lesser extent, microglia.[63] APOE normally functions to transport lipids, primarily cholesterol, to cells with APOE receptors such as neurons.[11,12,64] Inheritance of APOE4 has been linked to dysfunction in cholesterol efflux,[64] cerebral hypometabolism,[55] synaptic defects,[65] mitochondrial dysfunction,[66] blood-brain barrier dysfunction,[67] enhanced neuroinflammation,[68] cognitive impairment,[69] increased Aβ production and aggregation, decreased Aβ clearance and deposition,[70] neurotoxicity,[71] increased tau phosphorylation and oligomerization,[72] and tau-related neurofibril tangles.[73] APOE2 expression in an APOE4 background enhances Aβ clearance and reduces plaque formation, slowing down the progression of AD pathology.[24] Several teams investigated the effect of APOE2 on tau pathology and, while several potential mechanisms have been proposed, there is no conclusive impact or mechanism that has been identified.[47,71] In one suggested mechanism, APOE2 may interact with tau protein directly, leading to reduced tau phosphorylation and subsequent aggregation.[72,73] Another proposed mechanism suggests that APOE2 contributes to the preservation of tau phosphorylation signaling pathways, which in turn impact tau clearance.[24,74] These hypotheses are yet to be confirmed. In contrast, our data demonstrate that E2Ch increased tau clearance while E2 alone did not. The mechanisms by which this is accomplished, beyond just decreased heparin binding, remain to be elucidated.

APOE2 gene therapy for AD

The concept that CNS delivery of APOE2 could be therapeutic for APOE4 homozygotes with AD was first proposed by Kaplitt et al. in 1996.[32] Dodart et al.[33] provided support for this concept, using a lentivirus to delivery APOE2 to the hippocampus in PDAPP mice (overexpressing the V717F human APP mutation under the control of the platelet-derived growth factor promoter), resulting in decreased Aβ42 and Aβ40 levels compared to APOE3 and APOE4. Hudry et al.[75] used AAV4 to deliver APOE2 into the cerebral spinal fluid of mice overexpressing amyloid precursor protein demonstrating suppression of amyloid pathology. Together with the Paul laboratory,[48] we delivered APOE2 to the thalamus using the AAVrh.10 serotype in APOE4-associated amyloid mice, resulting in a reduction in amyloid pathology. Jackson and colleagues[35] used AAV delivery of ApoE2 to the ependymal cells lining the ventricles to suppress murine amyloid pathology. Alternative approaches to using APOE2 to treat APOE4 homozygotes have directly targeted suppression of APOE4 using oligonucleotides and microRNA[76] or by epigenetic silencing.[77] To date, all of these strategies have demonstrated suppression of amyloid pathology, but no prior studies have demonstrated positive impact of APOE variants on tau pathology.

Christchurch variant

The human Christchurch mutation was first identified in 1987 in a patient in New Zealand with type III hyperlipoproteinemia.[78] In 2019, Arboleda-Velasquez et al.[37] identified an individual with the Christchurch mutation on the APOE3 variant who had substantially delayed cognitive impairment despite a family history with the presenilin/E280A mutation that predisposes to AD. Positron emission tomography showed a high Aβ plaque burden, but the magnitude and spatial extent of tau burden were substantially lower than expected for individuals in the family with the presenilin/E280A variant. Studies of the APOE3Ch protein demonstrated decreased heparin binding compared to all other isoforms, suggesting the Ch variant may function to promote clearance of β-amyloid and decreases aggregation of extracellular tau.[37,46,79] That effect of Ch variant has been shown experimentally in human neurons and a mouse model of tauopathy, which provide further proof that the homozygous R136S mutation protects against APOE4-induced tau accumulation via the HSPG pathway.[80]

Gene therapy with APOE2Ch suppresses disease pathology in both amyloid and tau mouse models

Prior studies in mouse models from us and others[33,48,75] have demonstrated that APOE2 functions as a neuroprotective variant to treat APOE4-linked amyloid pathology. However, the data in the present study demonstrate APOE2 provides no therapeutic benefit in treating tau pathology. In contrast, genetic delivery to the CNS of the Ch variant of the APOE2 isoform suppresses not only amyloid pathology but also tau pathology.

Recent reports have outlined additional APOE-Christchurch mouse models, each incorporating the Christchurch variant into human APOE sequences, specifically human APOE3 and human APOE4.[79] (1) E3Ch treatment reduced tau tangles in APP/PS1 mice that had been administered human AD-tau.[80] (2) PS19 mice that are homozygous for Ch mutation reversed APOE4-dependent tau pathology, hippocampal shrinkage, and a reduction in microglial activity.[81] (3) In a PS19 mouse model on a hAPOE3 background, the Ch mutation provided protection from tau pathology and tau-induced synaptic impairments while suppressing interferon-related microglial activation. This is consistent with our studies that demonstrate amelioration of tau pathology with administered E2Ch to a tau mouse on an E4 background. These findings support the premise for our study that E2Ch provides protection against amyloid and tau pathologies. With the caveat that systemic Christchurch could mediate hyperlipidemia,[78] the CNS effects of Christchurch combined with E2 significantly improve the CNS amyloid and tau pathology associated with inheritance of the E4 allele.

Rationale for translation to a second-generation gene therapy for AD

Leveraging our clinical trial (NCT03634007 Gene Therapy for APOE4 Homozygote of Alzheimer's Disease) with the E2 construct administered to the cerebrospinal fluid (CSF), we anticipate that the intraparenchymal delivery of the small brain of the mouse will predict a similar protective outcome to human E4 homozygotes in the clinic. With that in mind, we show via our experiments that administration to the hippocampus leads to the biodistribution not only in the hippocampus but also in the cortex, midbrain, and in the cerebellum. The data show that APOE levels are elevated throughout the brain as expected for a secreted protein. Therefore, the hippocampal delivery in the mouse is proof of concept for translation to CSF delivery in the clinic.

Gene therapy to treat hereditary disorders conventionally delivers the normal allele to compensate for the loss of function caused by inherited mutations.[82] There are rare examples, however, when more effective gene therapy can be achieved using a gain-of-function human variant, such as the Padua variant used to treat factor IX hemophilia, which markedly improved efficacy at the same dose as that of the normal allele.[8,83] Although current knowledge for the use of gain-of-function mutations in gene therapy is limited, other beneficial mutations have been described, such as the CCR5 Delta 32 mutation, which provides HIV resistance.[84] Our study provides another example for the therapeutic use of gain-of-function variants, in this case for the treatment of AD. We propose that AAVrh.10hAPOE2Ch be considered as a therapy for individuals who are homozygous for APOE4 at high risk for the development of AD.

MATERIALS AND METHODS

The following is an overview of the methods; details can be found in supplemental information.

Heparin binding of APOE isoforms

Hemagglutinin (HA) 3′-tagged human (h) APOE variants hAPOE2, hAPOE2Ch, hAPOE3, hAPOE3Ch, and hAPOE4 were generated in

HEK293T cells by plasmid transfection. Assessment of heparin binding of the APOE isoforms was carried out as described by Futamura et al.[49]

AAV vectors

All vectors were based on the clade E nonhuman primate AAVrh.10 serotype, including AAVrh.10hAPOE2 (coding for the human APOE2 variant), AAVrh.10hAPOE2Ch (coding for human APOE2Christchurch), and AAVrh.10Null (as a control, identical but with no translatable sequence). The expression cassette for all the vectors uses the cytomegalovirus enhancer fused to the chicken β-actin promoter (CAG) promoter. For some experiments, a HA tag was added to the 3′ end of the coding sequence. The AAVrh.10 vectors were produced, purified, and characterized as previously described.[85]

Mouse models

APOE4-targeted replacement mouse (TRE4; mouse ApoE4 knockout, human APOE4 knockin) was obtained from P. Sullivan, Duke University Medical Center. Both the APP.PSEN1 (amyloid, B6C3-Tg/APPswe, PSEN1dE9/85#004462/034829 Mutant Mouse Resource and Research Center [MMRRC]), and P301S (tau, B6; C3-Tg/Prnp-MAPT*P301S/PS19Vle/J #008169) were obtained from Jackson Laboratories (Bar Harbor, ME). Both strains were crossed with TRE4 mice to generate the APP.PSEN1/TRE4 (humanized APOE4, amyloid mice) and P301S/TRE4 (humanized APOE4, tau mice) strains. Both strains were crossed to obtain homozygotes for APP.PSEN1 and P301S mutations. Genotypes were confirmed through analysis of tail snips or ear punches (Transnetyx, Cordova, TN). The APP.PSEN1/TRE4 mice were treated at 2.5 months and assessed at 5.5–6 months. The P301/TRE4 mice were treated at 5.5 months and assessed at 8.5– 9 months.

Vector administration and experimental groups

Intrahippocampal injection of AAV vectors was performed by stereotaxic surgery as previously described.[48] AAVrh.10Null, AAVrh.10hAPOE2, and AAVrh.10hAPOE2Ch vectors were administered at 4×10^{10} genome copies in 2 μL bilaterally via intrahippocampal route. Animals in PBS groups only administered in 2 μL of PBS with the same route. Amyloid mice were administered AAV vectors expressing E2-HA (8 male [M]/4 female [F]), E2Ch-HA (7M/5F), E2 (12M/10F), E2Ch (12M/11F), null (16M/10F), or PBS (15M/19F). Tau mice were administered AAV vectors expressing E2-HA (5M/4F), E2Ch-HA (5M/4F), E2 (5M/5F), E2Ch (8M/7F), null (10M/10F), or PBS (10M/10F).

Quantification of vector genome copies, transgene expression, and APOE protein levels

To quantify vector-related parameters, dissected brains were thawed and homogenized in ice-cold sterile Tris-buffered saline, pH 7.4 (TBS). Three equal volumes of homogenate were used for DNA, RNA, and protein isolation. DNA isolation was performed with a DNeasy tissue/blood DNA isolation kit, following the manufacturer's instructions (Qiagen, Valencia, CA). All DNA samples were quantified using the CMV-specific primer:probes forward 5′-GTCAATG GGTGGAGTATTTACGG-3′, reverse 5′-AGGTCATGTACTGGG CATAATGC-3′ by qPCR. RNA isolation was performed via Qiazol (Qiagen, Valencia, CA), and the final RNA was concentrated with RNeasy mini elute columns (Qiagen, Valencia, CA). DNA and RNA concentrations were determined with a Nanodrop (Thermo Fisher Scientific, Waltham, MA). All RNA samples were converted to cDNA with a high-capacity reverse transcription kit (Thermo Fisher Scientific). hAPOE mRNA levels were quantified using the primer/probes: forward 5′-GTGGAGAAGGTGCAGGCT-3′, reverse 5′-GATTGTCGCTGGGCACAGGG-3′ by RT-PCR and normalized to total RNA. Total protein content was determined by the Bradford assay.[86] APOE levels were measured with a human APOE-specific ELISA that equally recognizes each variant (ab108813, Abcam, UK) and normalized to total protein.

Quantification of insoluble and soluble Aβ42, Aβ40 levels

A sequential extraction method was used for the quantification of insoluble and soluble Aβ42 and Aβ40 peptides. TBS buffer containing 1% Triton X-100 plus guanidine HCl fractions together were considered as the insoluble fraction and the TBS fraction the soluble fraction. The total amount of Aβ42 and Aβ40 was assessed by ELISA and normalized to total protein.

Quantification of insoluble and soluble total tau and phospho-tau levels

Increased levels of phospho-tau, phosphorylated at residues such as Thr181, Thr231, and Ser396, are associated with neurofibrillary tangle formation and neuronal dysfunction.[24] Total tau levels reflect the overall tau burden and serve as a marker of neurodegeneration, as does tau solubility.[87] A sequential extraction method was used for the quantification of insoluble and soluble tau.

Histologic assessments

Following necropsy, one hemisphere of the mouse brain was fixed immediately in 4% paraformaldehyde in PBS, processed for embedding in paraffin, and serially sectioned at 5 μm in the sagittal plane (Histoserv, Germantown, MD). Sections identified by the gross morphology as the hippocampus were deparaffinized using xylene and a series of graded ethanol washes followed by treatment with a low-pH antigen-retrieval solution (00-4955-58, Thermo Fisher Scientific) at 88°C for 20 min. The sections were then blocked using the SuperBlock Buffer (37515, Thermo Fisher Scientific) overnight in a humid chamber at 4°C. The sections were then incubated at room temperature for 2 h with the primary and secondary antibodies.

MOAB-2

MOAB-2 antibody binds the N-terminal of β-amyloid peptides.[50] Staining was performed with MOAB-2 antibody (NBP2-13075, MOAB-2, Novus Biologicals).

GFAP

GFAP identifies astrocytes and was used for visualization and quantification of astrogliosis using anti-GFAP antibody (ab4674, Abcam).[53]

Iba-1

Iba-1 identifies microglia and was used for visualization and quantification of microglial activation using anti-Iba-1 antibody (019–19741, Fujifilm, Santa Clara, CA).[52]

AT8

AT8 recognizes total tau burden and the presence of neurofibrillary tangles and was used to assess these in tau mice (MN1020, AT8, Thermo Fisher Scientific).[54]

X-34

X-34 (SML1954, Sigma Aldrich) recognizes Aβ aggregates and was used to assess the total amyloid burden in amyloid mice.[51,88]

Quantification of histologic parameters

To quantify histologic parameters, sections for histology from tissue blocks that contained the hippocampus were selected. All were used but randomly assigned to the histological stain to avoid bias. The individuals performing the staining were blinded to the experimental conditions. Following staining, immunopositivity assessment was fully automated using QuPath software, with predefined settings and parameters to identify and quantify immunopositive cells in randomly selected hippocampal area. Staining intensity in X34 was measured with Haralick's texture features to assess the co-occurrence matrix in black background.[89] Utilizing grid-based sampling, we randomly selected starting points within the hippocampus and systematically sampled picture areas at regular intervals. This selection approach prevented bias and provided representative sampling across the region of interest.[90]

Behavioral assessments

Behavior was assessed over a period of 2 weeks in the amyloid mice ages 5.5–6 months, 3–3.5 months after therapy, and in the tau mice at age 8.5–9 months, 3–3.5 months after therapy. Neurological screening and behavioral deficits were evaluated with four behavioral tests during the experimental period, namely nest building, Y maze, novel-object recognition, and Barnes maze. All behavior equipment was cleaned with 70% ethanol between animals to avoid any olfactory cues in all behavioral tests except nesting. Behavior assays were performed on animals with blinded identification and data analysis utilizing fully automated software (AnyMaze, v7.0, 2022, Stoelting, San Diego, CA) to ensure unbiased evaluation without subjective input.[91] Nesting assays were an exception to the automated method, and nesting behavior assessment was scored by three or four blinded observers.

Nest building

Nest building is an innate behavior in mice and is considered an early indicator of behavioral deficits.[92] Two weeks before necropsies, animal bedding was changed with cotton squares and evaluated after 24 h by three or four blinded observers.[93] Animal nesting behavior was quantified by how the nesting material was used (using a five-point scale) and scored as follows: no nest-building behavior (score of 1), minimal engagement and organization in manipulating the nesting material and constructing the nest structure (score of 2), partial engagement and organization in manipulating the nesting material and constructing the nest structure (score of 3), near-complete nest structure (score of 4), and excellent performance characterized by meticulous gathering and arrangement resulting in a well-organized and stable nest structure (score of 5).[93]

Y maze

The Y maze measures spatial working memory and tests animals' innate curiosity to explore new environments and ability to differentiate between familiar and novel spatial cues.[57] Animals were placed on the Y-maze platform (Stoelting, IL) and recorded for 5 min. Consecutive novel arm entries were considered as alternations. Total arm entries and the number of alternations as assessed by the AnyMaze software were used to calculate the alternation ratio, a measure of novel exploratory behavior. To avoid possible effect of significant immobile time on alternation behavior, total number of entries was also assessed.

Novel-object recognition

The novel-object-recognition task assesses working memory and is based on the spontaneous tendency of mice to explore a novel object over a familiar one.[94] The novel-object-recognition task was performed on a novel-object-recognition test platform (Stoelting, IL) over 3 days, each pertaining to a unique phase. On day 1, the habituation phase, each animal was allowed to freely explore an open arena (40 × 40 × 30-cm tall white plastic box) for two 5-min trials with a 5-min inter-trial interval. On day 2, the familiarization phase, each animal was returned to the same arena for two 5-min sessions with two identical objects in opposite corners. After a 24-h retention period (the period to forget or remember object novelty), the testing phase was initiated on day 3, with the animal returning to the arena with two objects placed in positions identical to the previous day but with one replaced with a novel object. Mice were consistently placed in the arena facing the posterior wall to prevent object preference and allowed to explore for 5 min. The time spent exploring the novel objects divided by the time spent exploring both objects was calculated as the discrimination index. The criteria for exploration by the AnyMaze software were defined as the mouse sniffing, climbing on, or interacting while facing the novel object within 3 cm.

Barnes maze

The Barnes maze tests hippocampal-dependent spatial memory, capitalizing on the natural aversion of rodents to open spaces and their instinctual tendency to seek shelter.[95] The apparatus contains 20 circular holes (50-mm diameter) evenly spaced along the perimeter of the maze (Stoelting, IL). Only one hole provides access to an enclosed escape chamber (108 × 55 × 55 mm) beneath the maze floor. Testing procedures and distal visual cues were modified to enhance visuospatial learning and memory. Bright overhead lighting (10.8×10^2 lux) and white noise were used to drive escape motivation. The animals were trained for 7 days to use distal visual cues to locate the escape hole. The trial was terminated when the mouse had escaped through the hole or 180 s had elapsed. Latency to escape the maze

by entering the appropriate hole was calculated to assess spatial learning. Escape latency was evaluated in training days and probe trials. Data were recorded in a final trial performed 24 h after the last training day by closing the escape hole and recording latency using the 10- to 15-mm diameter setting in the AnyMaze software.

Statistical analyses

All experimental data were collected and analyzed using GraphPad Prism (GraphPad Software, La Jolla, CA). Linear quantitative data are presented as the mean ± standard error of means. Logarithmic scale data are presented as geometric mean ± geometric standard deviation. Data normality was assessed by Shapiro-Wilk's test. Based on data distribution, one-way ANOVA and Kruskal-Wallis tests were used. Multiple comparisons were performed with Dunn's and Bonferroni's *post hoc* analysis. Some experiments were carried out with HA-tagged vectors and some without the HA tag. Statistical differences between HA/non-HA and male/female results were analyzed by independent t test and Mann-Whitney U tests. There were no differences in the assessments with and without the tag, so the data were combined (Table S4). Likewise, there were no differences in the studies carried out with males vs. females, so data from both sexes were combined (Table S4). For all assessments, $p < 0.05$ was considered significant. With only two exceptions, the same number of animals were used for every assay in amyloid mice and, with the same two exceptions, the same number of animals were used for every assay in tau mice (Table S5).

DATA AND CODE AVAILABILITY

All data are available in the main text or the supplemental information.

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AUTHOR CONTRIBUTIONS

Conceptualization, C.G., D.S., S.M.K., and R.G.C.; methodology, C.G., D.S., S.M.K., and R.G.C.; software, C.G.; formal analysis, C.G., D.S., and P.L.L.; investigation, C.G. and H.L.; data curation management, C.G., D.S., S.M.K., and R.G.C.; writing – original draft, C.G., D.S., S.M.K., N.R.H., and R.G.C.; writing – review & editing, C.G., D.S., S.M.K., R.K., and R.G.C.; supervision, D.S., S.M.K., and R.G.C.; project administration, C.G., D.S., S.M.K., and R.G.C.; funding acquisition, R.G.C.

DECLARATION OF INTERESTS

D.S. and S.M.K. have equity in LEXEO Therapeutics; R.G.C. has equity and is a consultant to LEXEO Therapeutics; R.K. is an employee of LEXEO Therapeutics; R.G.C., C.G., D.S., and S.M.K. are named in a patent related to the subject of the study.

SUPPLEMENTAL INFORMATION

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